

Video image quality (new)

Per-frame Y-PSNR and SSIM from the recorded sample window. Source→stego includes H.264 encode + embedding; cover→stego isolates incremental embedding impact. The 40 dB mark is a reference quality floor, not part of the functional pass gate. Upscaled cases originate from CIF content.

Source	Resolution	Frames	Source PSNR μ	PSNR min	Source SSIM μ	SSIM min	Embed PSNR min	Embed SSIM min	Ref 40 dB	Pipeline
akiyo_cif.y4m	352x288	30	45.025	34.930	0.984862	0.950511	35.290	0.961177	NO	passed
akiyo_cif.y4m	640x480	30	46.317	33.510	0.987366	0.970163	33.680	0.980659	NO	passed
akiyo_cif.y4m	1280x960	30	48.282	40.810	0.990906	0.985761	41.550	0.993854	yes	passed
city_cif.y4m	352x288	30	43.089	38.900	0.986952	0.968763	40.850	0.979294	yes	passed
city_cif.y4m	640x480	30	44.421	37.830	0.989248	0.980642	38.870	0.990382	NO	passed
city_cif.y4m	1280x960	30	45.948	39.680	0.990852	0.987291	40.780	0.995957	yes	passed
coastguard_cif.y4m	352x288	30	42.848	37.140	0.987767	0.967239	38.400	0.977570	NO	passed
coastguard_cif.y4m	640x480	30	44.230	36.710	0.990017	0.979378	37.510	0.988349	NO	passed
coastguard_cif.y4m	1280x960	30	45.826	40.250	0.991254	0.987123	41.550	0.995537	yes	passed
container_cif.y4m	352x288	30	43.631	37.480	0.982994	0.952800	38.550	0.965819	NO	passed
container_cif.y4m	640x480	30	45.002	36.220	0.987409	0.973982	36.800	0.985115	NO	passed
container_cif.y4m	1280x960	30	46.716	40.830	0.990648	0.985831	42.020	0.994525	yes	passed
deadline_cif.y4m	352x288	30	43.770	35.670	0.987687	0.964626	36.310	0.973731	NO	passed
deadline_cif.y4m	640x480	30	45.169	39.220	0.989250	0.976537	40.380	0.985691	yes	passed
deadline_cif.y4m	1280x960	30	46.807	41.990	0.991250	0.986563	43.580	0.994567	yes	passed
football_cif.y4m	352x288	30	43.242	37.520	0.987327	0.970482	38.870	0.981025	NO	passed
football_cif.y4m	640x480	30	44.578	39.010	0.988966	0.981176	40.470	0.991175	yes	passed
football_cif.y4m	1280x960	30	46.195	41.880	0.990275	0.987232	43.970	0.996410	yes	passed
foreman_cif.y4m	352x288	30	53.349	41.800	0.998894	0.994378	42.100	0.995080	yes	passed
foreman_cif.y4m	640x480	30	51.818	40.990	0.997761	0.988720	41.300	0.990022	yes	passed
foreman_cif.y4m	1280x960	30	53.414	39.880	0.998132	0.992937	40.040	0.994210	yes	passed
foreman_cif_300f.y4m	352x288	30	53.392	43.070	0.998949	0.996039	43.430	0.996723	yes	passed

Video image quality (new) — continued

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Source	Resolution	Frames	Source PSNR μ	PSNR min	Source SSIM μ	SSIM min	Embed PSNR min	Embed SSIM min	Ref 40 dB	Pipeline
foreman_cif_300f.y4	640x480	30	51.758	39.180	0.997753	0.988477	39.420	0.989822	NO	passed
foreman_cif_300f.y4	1280x960	30	53.410	39.770	0.998129	0.992828	39.900	0.994037	NO	passed
hall_monitor_cif.y4	352x288	30	43.861	35.510	0.978761	0.945736	36.020	0.961932	NO	passed
hall_monitor_cif.y4	640x480	30	45.556	40.290	0.984355	0.970475	41.690	0.984258	yes	passed
hall_monitor_cif.y4	1280x960	30	47.062	32.650	0.989044	0.982073	32.760	0.991990	NO	passed

Per-frame quality artifact

Each frame's raw values and finite/infinite PSNR handling are in the CSV emitted with this media run.

Field	Meaning
Source→stego	Y-PSNR and SSIM relative to decoded Y4M at the output resolution; includes lossy encode
Cover→stego	Y-PSNR and SSIM relative to clean H.264 decode; isolates changes from embedding.
Sample	At most 30 initial frames per source/resolution; full-duration H.264 conversions are separ
Raw values	media_new\20260924T150707Z_2be60c\quality_per_frame_new.csv